

# Functions & the Haunted School

Pre-Assessment 4 · 11 questions · 40 points

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Score: / 40

This is a **pre-assessment**. It shows what you already know **before** we start this unit, so it is fine if some questions cover things you have not learned yet. Try every question you can. If you have no idea at all, write **“not yet”** and move on.

Work on your own, without a computer. You have about **45 minutes**. When a question asks what a program prints, write exactly what Python would print, one line of output per answer line. If the code would crash, write the name of the error (for example, `TypeError`). Show your thinking; partial credit is given.

## 1. Function vocabulary

2 points

```
def greet(name):  
    print("Hello, " + name)  
  
greet("Ana")
```

Use a word from this bank: **parameter, argument, return value, docstring**.

(a) In this program, `name` is a \_\_\_\_\_. (1 pt)

\_\_\_\_\_

(b) In this program, `"Ana"` is an \_\_\_\_\_. (1 pt)

\_\_\_\_\_

## 2. Cheer Machine

3 points

Write exactly what this program prints.

```
print("Start")  
  
def cheer(team):  
    print("Go " + team + "!")  
  
cheer("Tigers")  
cheer("Owls")  
print("End")
```

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### 3. Order Up

4 points

```
def order(food, size="medium", drink="water"):
    print(f"{size} {food} with {drink}")
```

Write what each call prints.

| Call                               | Prints |
|------------------------------------|--------|
| order("pizza")                     |        |
| order("tacos", "large")            |        |
| order("soup", drink="juice")       |        |
| order(drink="milk", food="cereal") |        |

### 4. Return or print?

4 points

```
def double(n):
    return n * 2

def show_double(n):
    print(n * 2)

a = double(5)
b = show_double(5)
print(a)
print(b)
```

(a) Write exactly what the program prints.

(3 pts)

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(b) Why is `b` different from `a`?

(1 pt)

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## 5. Even or Odd?

3 points

```
def is_even(n):  
    return _____
```

(a) Fill in the blank so `is_even` returns `True` for even numbers and `False` for odd numbers. (2 pts)

(b) With your function, what does `is_even(7)` return? (1 pt)

## 6. Inside and outside

4 points

(a) What does this print? (2 pts)

```
score = 10  
  
def add_bonus():  
    score = 50  
    print("inside:", score)  
  
add_bonus()  
print("outside:", score)
```

(b) What happens when this runs, and why? (2 pts)

```
def make_name():  
    nickname = "Ace"  
  
make_name()  
print(nickname)
```

## 7. Function Detective

3 points

```
def average(a, b):  
    """Return the average of two numbers."""  
    return a + b / 2
```

(a) What is the line in triple quotes called?

(1 pt)

(b) A test says `average(4, 6)` should be 5. What does the function actually return?

(1 pt)

(c) Rewrite the `return` line so the test passes.

(1 pt)

## 8. Long names

3 points

```
def count_long(words, n):  
    total = 0  
    for w in words:  
        if len(w) > n:  
            total = total + 1  
    return total  
  
pets = ["cat", "rabbit", "dog", "hamster", "fish"]  
print(count_long(pets, 3))  
print(count_long(pets, 5))  
print(count_long(pets, 10))
```

Write exactly what the program prints.

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## 9. Refactor Day

4 points

This code repeats itself. Write a function `announce(name, game)` that prints the two lines for one player, then write the calls that make the program print exactly the same thing.

```
print("==== Ana ====")
print("Ana, it's your turn to play chess!")
print("==== Ben ====")
print("Ben, it's your turn to play cards!")
print("==== Cy ====")
print("Cy, it's your turn to play chess!")
```

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## 10. Secret Initials

4 points

Write a function `make_initials(full_name)` that **returns** the initials in capital letters, each followed by a dot. For example, `make_initials("ada lovelace")` returns `"A.L."` and `make_initials("grace brewster hopper")` returns `"G.B.H."`.

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## 11. Escape the Haunted School

6 points

Here is part of a text adventure.

```
rooms = {
    "hall": {"north": "library", "east": "gym"},
    "library": {"south": "hall"},
    "gym": {"west": "hall", "north": "lab"},
    "lab": {"south": "gym"},
}

def move(current, direction):
    if direction in rooms[current]:
        return rooms[current][direction]
    print("You can't go that way.")
    return current

room = "hall"
for d in ["east", "north", "west", "south"]:
    room = move(room, d)
    print(room)
```

(a) Write exactly what the program prints.

(3 pts)

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(b) Write a function `has_won(inventory, room)` that returns `True` when the player is in the "lab" **and** has "key" in the `inventory` list, and `False` otherwise.

(3 pts)

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**END OF ASSESSMENT**